

Branch Scaling for the Square-Billiard Abel Trace

Route-A structural certificate C162: baseline manuscript

Abstract

Starting from the exact Poisson representation of the unit-square Dirichlet Abel half-wave trace, the baseline manuscript identifies the natural $\epsilon^{3/2}$ scaling of one matching dual-lattice summand. It computes the principal-branch coefficient at a shell time. Passage through the complete infinite trace, coincident simple poles, and negative-time conjugation remain explicitly open at this stage.

Keywords: square billiard; Abel trace; branch singularity; local scaling; clean family.

中文摘要：基线稿从正方形狄利克雷阿贝尔半波迹的精确泊松表示出发，确定单个匹配对偶格点项的 $\epsilon^{3/2}$ 分支尺度，并计算主值分支系数。如何穿过完整无限迹、处理重合简单极点及负时间极限，在本轮仍明确保留为开放问题。

关键词：正方形台球；阿贝尔迹；分支奇点；局部尺度；光滑轨道族。

1 Local branch calculation

For $\Re s > 0$,

$$W_D(s) = \frac{s}{2\pi} \sum_{m \in \mathbb{Z}^2} (s^2 + 4|m|^2)^{-3/2} - \frac{1}{4} - \frac{1}{e^{\pi s} - 1}.$$

At $t_N = 2\sqrt{N}$, a matching vector satisfies $s^2 + 4N = \epsilon(\epsilon - 2it_N)$ for $s = \epsilon - it_N$. Therefore

$$\lim_{\epsilon \downarrow 0} \epsilon^{3/2} \frac{s}{2\pi} (s^2 + 4N)^{-3/2} = \frac{e^{i\pi/4}}{8\pi N^{1/4}}.$$

The phase is fixed by the principal power. This establishes the correct local scale but, at baseline, does not interchange the limit with the infinite sum.

2 Baseline boundary

The source objects are clean lattice families rather than isolated primitive orbits. No isolated stability amplitude, target divisor law, arithmetic Euler factor, Hilbert–Pólya construction, or Route B is claimed.

Declarations

All source data and code are packaged. No external dataset, subjects, competing interest, or external funding is involved. An AI coding assistant supported drafting and code review but was not an external reviewer.